

Yifan Zhang

Department of Computer Science
Missouri State University, MO, 65897
Phone: +1 (775) 200-3895
Email: yifanzhang@missouristate.edu
LinkedIn: [linkedin.com/in/yifan-zhang](https://www.linkedin.com/in/yifan-zhang)
Google Scholar: scholar.google.com/yifan-zhang

RESEARCH INTERESTS

Deep Learning: supervised learning, time series forecasting, image restoration.
Bioinformatics: unsupervised learning, single-cell RNA sequencing data analysis.

EDUCATION

University of Nevada, Reno <i>Ph.D. Computer Science and Engineering</i> - Dissertation Title: Advancing Time Series Forecasting: Innovative Machine Learning Approaches - Research Areas: Deep Learning, Data Mining, Multivariate Time Series Forecasting - Co-advisors: Dr. Frederick C Harris, Jr. and Dr. Rui Wu (ECU)	Reno, NV, USA Jan. 2020 – August 2024
Nanjing University of Aeronautics and Astronautics <i>M.Eng. Control Science and Engineering</i> - Thesis Title: Research on Image Super-Resolution Algorithm Based on Deep Learning - Research Areas: Deep Learning, Image Super-Resolution - Advisor: Dr. Xin Yang	Nanjing, Jiangsu, China Sep. 2016 – April 2019
Huainan Normal University <i>B.Eng. Automation</i>	Huainan, Anhui, China Sep. 2012 – June 2016

EXPERIENCE

Assistant Professor <i>Missouri State University</i>	Aug. 2024 – Present Springfield, MO
Research Assistant <i>University of Nevada, Reno</i>	Jan. 2020 – July 2024 Reno, NV
NSF REU Site Mentor <i>University of Nevada, Reno</i>	June 2023 – July 2023 Reno, NV
Software Engineer <i>ZTE Corporation</i>	Apr. 2019 – July 2019 Nanjing, China
Visiting Student <i>Multimedia Lab, Shenzhen Institutes of Advanced Technology (SIAT)</i>	Nov. 2017 – Mar. 2018 Shenzhen, China
Research Assistant <i>Nanjing University of Aeronautics and Astronautics (NUAA)</i>	Sep. 2016 – Mar. 2019 Nanjing, China

TEACHING EXPERIENCE

Guest Lecture <i>University of Nevada, Reno</i> CS 135 Computer Science I: C Programming Language	Aug. 2022 - May 2024
Teaching Assistant <i>University of Nevada, Reno</i> CS 135 Computer Science I: C Programming Language	Aug. 2022 - May 2024
Teaching Assistant <i>University of Nevada, Reno</i> CS 425 Software Engineering & CS 426 Senior Projects	Jan. 2020 - May 2022
Teaching Assistant <i>Nanjing University of Aeronautics and Astronautics</i> 7D031022L: Introduction to Artificial Intelligence	Sep. 2016 – Jan. 2017

Time Series Forecasting | *University of Nevada, Reno*

July. 2020 – present

- Dozerformer: A sequence adaptive sparse transformer for MTS forecasting. It comprises Local, Stride, and Vary components designed to capture the locality, seasonality, and global temporal dependencies of time series data. Notably, the Vary component dynamically increases historical time steps utilized as the forecasting horizon extends. Those components significantly enhance the efficiency of the transformer, transitioning it from a quadratic to a linear computational complexity.
Software: <https://github.com/GRYGY1215/Dozerformer>
- MTPNet: We developed a novel Multi-scale Transformer Pyramid Network (MTPNet), specifically designed to effectively capture temporal dependencies at multiple unconstrained scales. The predictions are inferred from multi-scale latent representations obtained from transformers at various scales.
Software: <https://github.com/GRYGY1215/MTPNet>
- eGRU: A novel extreme event adaptive Gated Recurrent Unit (eGRU) was introduced. It employs two hidden states to discern and learn patterns from both normal and extreme event data within time series data.
Software: <https://github.com/GRYGY1215/eGRU>
- DRFEE: We proposed a framework that addresses extreme events within time series data to improve the accuracy of water level predictions. It utilizes two distinct machine learning models to generate predictions for normal and extreme events, respectively.

scRNA-seq Trajectory Inference | *University of Nevada, Reno*

Jan. 2021 – Nov. 2022

- We developed a robust and accurate single-cell data trajectory inference method using ensemble pseudotime. This method includes pathway analysis, which groups genes based on pathways to generate factors. The ensemble pseudotime inference algorithm leverages multiple clustering results, generating a more robust pseudotime. In this method, the trajectory determined by the minimum spanning tree undergoes refinement by sorting its nodes based on their average pseudotime.
Software: <https://cran.r-project.org/package=scTEP>

Robust Multi-Image Adaptive Super-Resolution | *NCAA*

Sep. 2016 – Mar. 2019

- We proposed a densely connected feature extraction module based on CNNs. Multiple feature extraction modules are densely connected to generate hierarchical feature maps. Subsequently, Subpixel Convolutional Neural Networks are employed for up-sampling the image resolution.

Deep Learning for Image Restoration | *SIAT*

Nov. 2017 – Mar. 2018

- We proposed a CNN-based iterative module comprising both an up-sampling path and a down-sampling path. This module is designed to generate high-resolution and low-resolution images. We introduced a framework that iteratively executes the proposed iterative module multiple times, generating intermediate low- and high-resolution images. The final high-resolution image output is computed as the weighted sum of all intermediate high-resolution images. The loss function comprises the L2 norm for both low-resolution and high-resolution images.

PUBLICATIONS**Journals**

- [J1] **Zhang Y**, Wu R, Dascalu S, and Harris, Jr. F, Dozerformer: Sequence Adaptive Sparse Transformer for Multivariate Time Series Forecasting. *Nature Scientific Reports*, 14, 15909 (2024). <https://doi.org/10.1038/s41598-024-66886-1>
- [J2] **Zhang, Y.**, Wu, R., Dascalu, S.M. et al. A novel extreme adaptive GRU for multivariate time series forecasting. *Nature Scientific Reports* 14, 2991 (2024). <https://doi.org/10.1038/s41598-024-53460-y>
- [J3] **Y. Zhang**, R. Wu, S. M. Dascalu and F. C. Harris, "Multi-Scale Transformer Pyramid Networks for Multivariate Time Series Forecasting," in *IEEE Access*, vol. 12, pp. 14731-14741, 2024, doi: 10.1109/ACCESS.2024.3357693.
- [J4] **Zhang, Y.**, Tran, D., Nguyen, T., et al. A robust and accurate single-cell data trajectory inference method using ensemble pseudotime. *BMC Bioinformatics* 24, 55 (2023). <https://doi.org/10.1186/s12859-023-05179-2>
- [J5] Yang, X., **Zhang, Y.**, Li, T. et al. Image Super-Resolution Based on the Down-Sampling Iterative Module and Deep CNN. *Circuits Syst Signal Process* 40, 3437–3455 (2021). <https://doi.org/10.1007/s00034-020-01630-4>
- [J6] Yang, X., **Zhang, Y.**, Guo, Y. et al. An image super-resolution deep learning network based on multi-level feature extraction module. *Multimed Tools Appl* 80, 7063–7075 (2021). <https://doi.org/10.1007/s11042-020-09958-4>

[J7] Yang, X., Zhu, S., Zhou, D., **Zhang Y.**, An improved target tracking algorithm based on spatio-temporal context under occlusions. *Multidim Syst Sign Process* 31, 329–344 (2020). <https://doi.org/10.1007/s11045-019-00664-5>

Conferences

[C1] **Zhang Y**, Wu R, Dascalu S, and Harris, Jr. F, A User Study of Two Downstream Single-cell Data Analysis Methods: Clustering and Trajectory Inference. 21st International Conference on Information Technology-New Generations 2024

[C2] Cardoza-Aguila J, Milbourn C, **Zhang Y**, Yang L, Wu R, Dascalu S, and Harris, Jr. F. A Holistic Approach for single-cell Data Trajectory Inference using Chromosome Physical Location and Ensemble Random Walk. 21st International Conference on Information Technology-New Generations 2024

[C3] Lewis C, Enriquez D, Calabrese L, **Zhang Y**, Anbro SJ, Houmanfar RA, Crosswell LH, Rebaleati MJ, Starmer LA, Harris. Jr FC. Virtual Reality Multiplayer Interaction and Medical Patient Handoff Training and Assessment. In ITNG 2022 19th International Conference on Information Technology-New Generations 2022 Feb 14 (pp. 17-23). Springer International Publishing.

[C4] Cuellar A, **Zhang Y**, Dascalu SM, Harris Jr FC. Social Media User Study 37. In ITNG 2022 19th International Conference on Information Technology-New Generations 2022 (Vol. 1421, pp. 303-309). Springer Nature.

[C5] **Yifan Zhang**, Jiahao Li, Ablan Carlo, Alex K Manda, Scott Hamshaw, Sergiu M. Dascalu, Frederick C. Harris, and Rui Wu. Data regression framework for time series data with extreme events. In 2021 IEEE International Conference on Big Data (Big Data), pages 5327–5336, 2021. doi:10.1109/BigData52589.2021.9671387.

ACADEMIC ADVISING

Jovany Cardoza-Aguilar Undergraduate student, <i>University of Washington Tacoma</i>	June 2023 - Oct. 2023
Caleb Milbourn Undergraduate student, <i>University of Houston</i>	June 2023 - Oct. 2023
Augustine Ofoegbu Master's student, <i>University of Nevada, Reno</i>	Aug. 2023 - present
Grant Melvin Master's student, <i>East Carolina University</i>	Jan. 2024 - present

ACADEMIC SERVICES

Journals Reviewer:

- IEEE Internet of Things
- China Communications
- PLOS Computational Biology
- Frontiers in Environmental Science

Conferences Reviewer:

- The Thirteenth International Conference on Learning Representations (ICLR 2025)
- 31st International Conference on Software Engineering and Data Engineering
- 38th International Conference on Computers and Their Applications

Student Volunteer:

- 2021 IEEE International Conference on Big Data (Big Data)

HONORS AND AWARDS

IEEE BigData 2021 Student Travel Award, *IEEE BigData-21*, \$300, 2021
GSA Travel Award, *University of Nevada, Reno*, \$300, 2021
Advanced Individual in Social Activities Award, *NUAA*, 2017
First-class Graduate Academic Scholarship, *NUAA*, \$1,500, 2016, 2017, 2018
National Scholarship for Graduate Students, *NUAA*, \$900, 2016, 2017, 2018

SKILLS

Programming Languages: Python, C/C++, R

Frameworks: MATLAB, Proteus, AutoCAD

Libraries: Pandas, Numpy, Tensorflow, PyTorch, Matplotlib

Languages: Chinese, English